	Case Name: Apartment Building (4)	Sector	Construction (residential building)	
OR-AS Operations Research Applications and Solutions	**OR-AS** Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
			One of nine std. scenarios	
Submitted by	Nicolas Van Den Noortgate		User defined distributions	
Date	2014	Project Control	Automatic tracking	
File Name	C2014-08 Apartment Building (4)		Tracking based on user input	

1. Project description

Project authenticity

The project involves the renovation and technical upgrade of a multi-storey residential building, including demolition works, structural adjustments, installation of utilities, interior finishing across six floors and the ground floor, as well as roof insulation, lift repairs, and final clean-up.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	42
Planned Duration (PD)	233
Budget At Completion (BAC)	1,992,222.09 €
Renewable Resources	-
Consumable Resources	-

Network topology	
Serial/Parallel (SP)	44%
Activity Distribution (AD)	29%
Length of Arcs (LA)	11%
Topological Float (TF)	14%

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	14	10	1.4
CRI-rho	14	10	1.44
CRI-tau	8	6	1.72

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	46	46	0.15
SI	20	34	1.57
SSI	10	13	1.39
CRI-r	13	12	1.43
CRI-rho	12	12	1.62
CRI-tau	8	8	1.82

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	6.53	1.15
PV - SPI	28.91	26.19
PV - SCI	29.62	26.82
ED - 1	24.45	21
ED - SPI	28.91	26.19
ED - SCI	29.15	26.36
ES - 1	2.34	-1.36
ES - SPI(t)	12.09	9.66
ES - SCI(t)	12.6	10.02

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	1.11	-0.44
CPI	2.13	0.17
SPI	22.15	21.73
SPI(t)	12.51	12.07
SCI	22.83	22
SCI(t)	13.3	12.46
0.8 CPI + 0.2 SPI	9.79	9.24
0.8 CPI + 0.2 SPI(t)	4	3.41

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 13 tracking. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-165894.13	-133144.05	-17.79	0.85	0.86	0.89	0.99
std dev	147217.86	122654.23	12.87	0.05	0.08	0.07	0.01
final	-388078.25	0	-42	0.84	1	0.85	1

2.3.3.2. Time forecasting

PD	233

Real Duration	275

Late/early	18.03 %

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	249	14.42	6.87	-6.87
PV - SPI	272.85	26.11	17.1	-17.1
PV - SCI	323.31	33.22	38.76	-38.76
ED - 1	257.08	13.34	10.33	-10.33
ED - SPI	278	20.99	19.31	-19.31
ED - SCI	303.08	34.28	30.08	-30.08
ES - 1	251.23	13.31	7.82	-7.82
ES - SPI(t)	264.23	23.31	13.41	-13.41
ES - SCI(t)	287.54	40.42	23.41	-23.41

2.3.3.3. Cost forecasting

BAC	1,992,222.09 €

Real Cost	2,380,299.86 €

Over/under budget	19.48 %

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	2158116.13	147217.87	9.33	9.33
CPI	2363779.52	140755.91	3.88	0.69
SPI	2362878.99	152038.8	3.69	0.73
SPI(t)	2287284.35	202434.69	6.68	3.91
SCI	2606063.92	295488.34	9.61	-9.49
SCI(t)	2521344.06	348464.69	7.46	-5.93
0.8 CPI + 0.2 SPI	2360038.15	114080.29	2.97	0.85
0.8 CPI + 0.2 SPI(t)	2344938.48	128682.43	3.44	1.48